SIMATS ENGINEERING

ASSESSMENT TOOL 2: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**R. Pranathi, M. Meghana**
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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 5

Code of Conduct

We R. Pranathi (192525076), M. Meghana(192525435) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pranathi
Meghana

Assessment Tasks

Technical Presentation Topics Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

1. Cloud computing characteristics and benefits for startup ecosystems.
2. How virtualization enabled the growth of cloud service providers.
3. A comparative presentation on IaaS, PaaS, and SaaS for an educational institution.
4. Web services in cloud platforms: SOAP vs REST for enterprise integration.
5. Why XaaS is becoming a dominant consumption model in modern IT.

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

CO1-AT2-TECHNICAL PRESENTATION

CSA1522 : Cloud Computing for Bigdata Analytics

PRESENTED BY :
R. Pranathi(192525076)
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How virtualization enabled the growth of cloud service providers

What is Virtualization?

Definition

- Virtualization is the creation of multiple virtual machines (VMs) on a single physical server.
- Each VM has its own operating system and applications.
- It allows efficient use of hardware resources.

Key Components:

- Physical Server
- Hypervisor
- Virtual Machines (VMs)

What is Cloud Computing?

Cloud Computing

- Delivers computing services over the internet.
- Users can access servers, storage, databases, and software on demand.

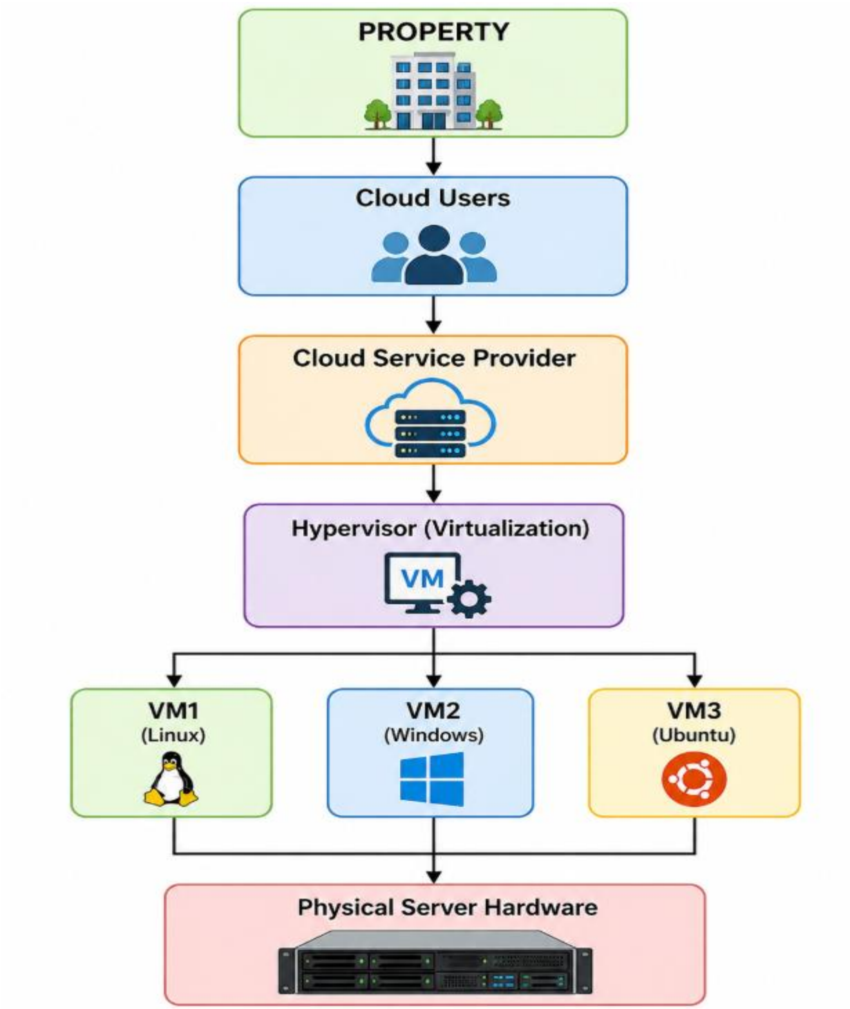
Cloud Service Models:

- IaaS
- PaaS
- SaaS

How Virtualization Works

- **Working Process**
- Physical server is installed.
- Hypervisor is installed.
- Hypervisor creates multiple VMs.
- Each VM runs independently.
- Users access services through the cloud.

Architecture Diagram



Benefits of Virtualization

Advantages

- Better resource utilization
- Reduced hardware cost
- Easy scalability
- Faster deployment
- High availability
- Disaster recovery
- Lower power consumption
- Improved security through VM isolation

Real-World Use Case

Amazon Web Services (AWS EC2)

- Uses virtualization to create virtual servers.
- Customers launch VMs within minutes.
- Pay only for used resources.
- Supports businesses worldwide.

Other Providers:

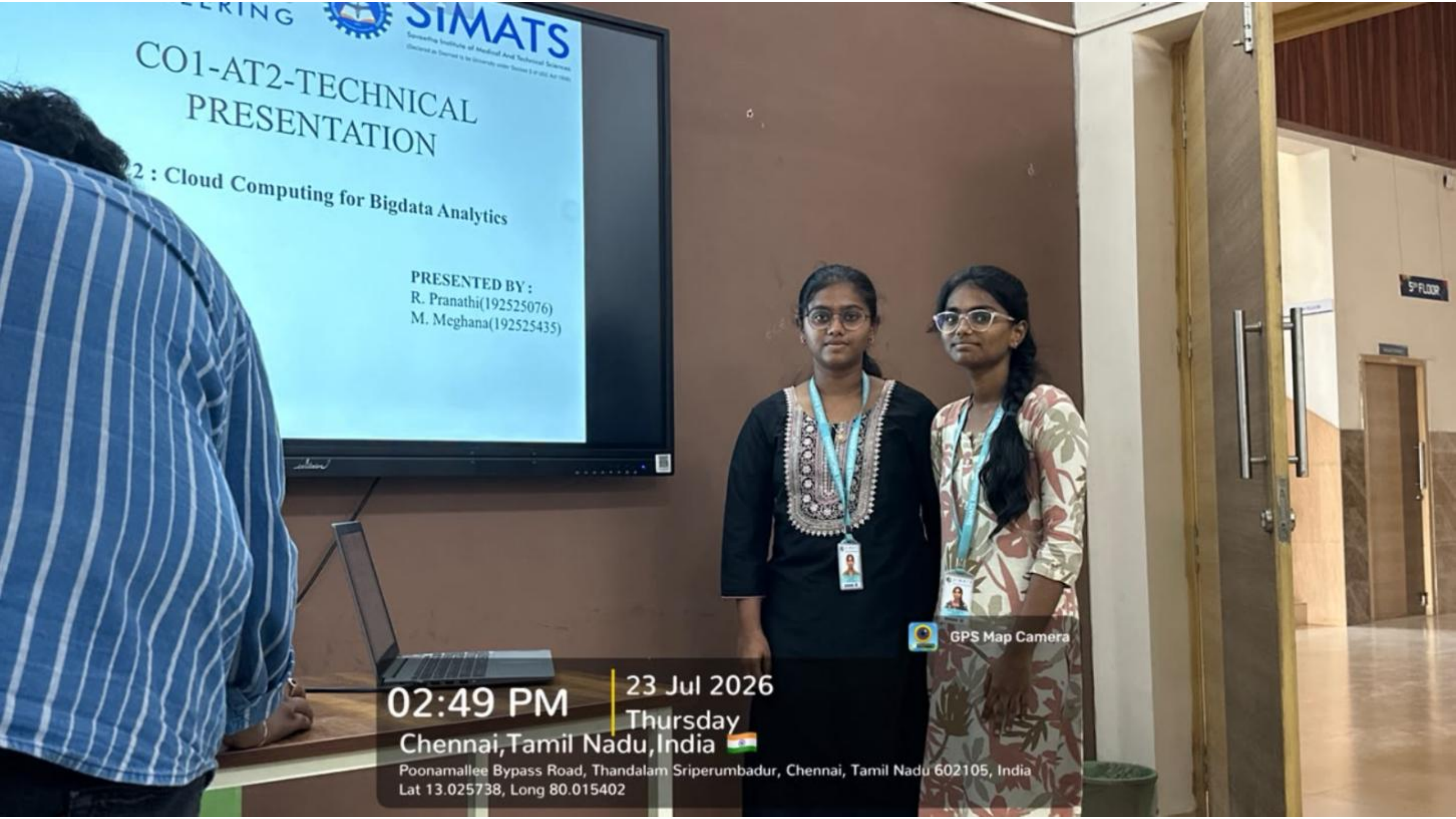
- Microsoft Azure
- Google Cloud Platform
- IBM Cloud

Why Virtualization Helped Cloud Providers Grow

- Reduced infrastructure costs.
- Increased server utilization.
- Enabled on-demand resource allocation.
- Supported millions of users simultaneously.
- Improved reliability and scalability.
- Made cloud services affordable.

Conclusion

- Virtualization is the foundation of cloud computing.
- It allows multiple virtual machines to share one physical server.
- Cloud providers can deliver scalable, flexible, and cost-effective services.
- Modern cloud platforms rely heavily on virtualization technology.


SIMATS
CO1-AT2-TECHNICAL PRESENTATION
2 : Cloud Computing for Bigdata Analytics
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23 Jul 2026
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Thank you